

2024 Higher Chemistry Paper 2 - Q6

Section: Chemistry in Society

Topic: Controlling the Rate (Biodiesel, Esters)

Question summary (Q6):

Biodiesel is made from plant oils by trans-esterification:

oil + methanol $\rightarrow$ biodiesel + glycerol.

(a) (i) Name the ester $\text{CH}_3(\text{CH}_2)_6\text{COOCH}_3$.

(a) (ii) Draw and label apparatus suitable for preparing an ester safely in the lab.

Worked Solution:

(a) (i) $\text{CH}_3(\text{CH}_2)_6\text{COOCH}_3$:

- Parent acid: octanoic acid ($\text{CH}_3(\text{CH}_2)_6\text{COOH}$).
- Alcohol: methanol ($-\text{OCH}_3$).
- Ester name: methyl octanoate.

(a) (ii) Apparatus for ester preparation:

- Use reflux apparatus to heat the carboxylic acid + alcohol with catalyst (conc. H_2SO_4) without losing volatile vapours.
- Setup includes: round-bottom flask, reflux condenser (water in at bottom, out at top), heat source (water bath or heating mantle), and anti-bumping granules.
- This ensures safe heating of volatile, flammable reactants.

Final Answer:

(a) (i) Methyl octanoate

(a) (ii) Reflux setup with round-bottom flask, condenser, heat source

Revision Tips:

- Esterification = carboxylic acid + alcohol $\rightarrow$ ester + water.
- Lab prep requires reflux to prevent loss of volatile reactants/products.
- Esters: sweet smells, used in perfumes, flavourings, solvents,

biodiesel.